PATIENT INFORMATION	SPECIMEN INFORMATION	PROVIDER INFORMATION
LAST, First ID#: DOB: Month, Day, Year Sex:	Type: Collected: Month, Day, Year Received: Month, Day, Year PG ID: ####-###-### Test Method: PGxome	Physician Genetic Counselor Institution

PGxome - Whole Exome Sequencing

Singleton Analysis (Proband Only)

PRIMARY FINDINGS: Heterozygous for a Likely Pathogenic Variant in *ZEB2*

SECONDARY FINDINGS: Heterozygous for a Likely Pathogenic Variant in *NOD2*

CARRIER STATUS: Heterozygous for a Pathogenic Variant in *DPYS* and a Likely Pathogenic Variant in *GJB2*

INDICATION FOR TESTING: global developmental delay, aortic valve stenosis (mild), mild aortic stenosis, Wide anterior fontanel, febrile seizures, patent ductus arteriosus, atopic dermatitis, Mongolian blue spot, Abnormal number of hair whorls (two parietal), midface retrusion, hypertelorism, prominent epicanthal folds, central hypotonia

Variants in genes known to be associated with phenotype:

Sequence Variant(s):

Gene, Transcript	Mode of Inheritance, Gene OMIM	DNA Variations, Predicted Effects, Zygosity	ClinVar ID	Highest Allele Frequency in a gnomAD Population	In Silico Missense Predictions	Interpretation
ZEB2, NM_014795.3	AD, 605802	c.1349_1350del, p.Phe450Serfs*5, Heterozygous	Not listed in ClinVar	Not Present	Not Applicable	LIKELY PATHOGENIC

Mode of Inheritance: Autosomal Dominant=AD, Autosomal Recessive=AR, X-Linked=XL
ClinVar ID: Variant accession (www.ncbi.nlm.nih.gov/clinvar)
GnomAD: Allele Frequency registered in a large population database (gnomad.broadinstitute.org). Value listed is the highest allele frequency reported within one of seven population categories recognized in gnomAD v.2.1.1 (The “Other” population is excluded).
Missense Predictions: Summarized output (Damaging, Conflicting, or Tolerated) via PolyPhen-2, SIFT, MutationTaster, and FATHMM (PMID: 26555599).

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***ZEB2* VARIANT INFORMATION:**

This patient is heterozygous in the *ZEB2* gene for a variant designated c.1349_1350del, which is predicted to result in a frameshift and premature protein termination (p.Phe450Serfs*5). To our knowledge, this variant has not been reported in the literature or in a large population database (http://gnomad.broadinstitute.org), indicating this variant is rare. Frameshift variants in *ZEB2* are expected to be pathogenic. This variant is interpreted as likely pathogenic.

Pathogenic variants in *ZEB2* are associated with autosomal dominant Mowat-Wilson syndrome (OMIM #235730), which is characterized by intellectual disability, gross developmental delay, epilepsy, and diverse congenital anomalies. The majority of pathogenic variants described are nonsense and frameshift alterations.

Variants in genes possibly associated with the phenotype: None detected

ACMG Guideline Medically Actionable Gene Variants*: None detected

*Recommended list of genes (Miller et al. 2022. PubMed ID: 35802134)

Variants in genes known to be associated with phenotype:

Sequence Variant(s):

Gene, Transcript	**Mode of Inheritance, Gene OMIM**	**DNA Variations, Predicted Effects, Zygosity**	**ClinVar ID**	**Highest Allele Frequency in a gnomAD Population**	**In Silico Missense Predictions**	**Interpretation**
NOD2, NM_022162.2	AD, 605956	c.855del, p.Gly286Alafs*91, Heterozygous	Not listed in ClinVar	Not Present	Not Applicable	LIKELY PATHOGENIC

*For footnotes, please see previous table.

***NOD2* VARIANT INFORMATION:**

This patient is heterozygous in the *NOD2* gene for a variant designated c.855del, which is predicted to result in a frameshift and premature protein termination (p.Gly286Alafs*91). To our knowledge, this variant has not been reported in the literature or in a large population database (http://gnomad.broadinstitute.org), indicating this variant is rare. Frameshift variants in *NOD2* are expected to be pathogenic. This variant is interpreted as likely pathogenic.

Pathogenic variants in *NOD2* have been associated with multifactorial Inflammatory bowel disease 1, Crohn disease (OMIM #266600), susceptibility to psoriatic arthritis (OMIM #607507), multifactorial Yao syndrome (OMIM #617321), and autosomal dominant Blau syndrome (OMIM #186580).

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Carrier status for autosomal and X-linked recessive disorders:

Sequence Variant(s):

Gene, Transcript	Mode of Inheritance, Gene OMIM	DNA Variations, Predicted Effects, Zygosity	ClinVar ID	Highest Allele Frequency in a gnomAD Population	In Silico Missense Predictions	Interpretation
DPYS, NM_001385.2	AR, 613326	c.1393C>T, p.Arg465*, Heterozygous	863103	0.08%, Eastern Asian	Not Applicable	PATHOGENIC
GJB2, NM_004004.5	AR, 121011	c.109G>A, p.Val37Ile, Heterozygous	Not listed in ClinVar	8.2%, Eastern Asian	Conflicting	LIKELY PATHOGENIC

*For footnotes, please see previous table.

***DPYS* VARIANT INFORMATION:**
This patient is heterozygous in the *DPYS* gene for a variant designated c.1393C>T, which is predicted to result in a premature protein termination (p.Arg465*). This variant was reported in an individual with Dihydropyrimidinase deficiency (van Kuilenburg et al 2010. PubMed ID: 20362666; Hiratsuka M et al 2015. PubMed ID: 25915935; Hishinuma E et al 2017. PubMed ID: 28642038; Hou YC et al 2020. PubMed ID: 31980526; Placeholder for https://www.ncbi.nlm.nih.gov/pmc/?term=4544753). This variant is reported in 0.085% of alleles in individuals of East Asian descent in gnomAD (http://gnomad.broadinstitute.org/variant/8-105405062-G-A). Nonsense variants in *DPYS* are expected to be pathogenic. This variant is interpreted as pathogenic.

Pathogenic variants in *DPYS* have been associated with autosomal recessive dihydropyrimidinuria (OMIM #222748).

***GJB2* VARIANT INFORMATION:**
This patient is also heterozygous in the *GJB2* gene for a variant designated c.109G>A, which is predicted to result in the amino acid substitution p.Val37Ile. This variant in the homozygous state has been reported to be strongly associated with both mild-to-moderate (P=2.0×10-11) and severe-to-profound (P=0.001) hearing impairment, but was estimated to have a low penetrance (17%). Onset of hearing loss was 65% congenital and 35% delayed in individuals that were homozygous for c.109G>A (Chai et al. 2015. PubMed ID: 24654934). This variant has also been reported to cause nonsyndromic hearing loss in the compound heterozygous state with a second disease causing *GJB2* variant (Abe et al. 2000. PubMed ID: 10633133). The c.109G>A variant has been reported at a subpopulation frequency of up to 8.2% in a database of individuals with unknown phenotype (http://gnomad.broadinstitute.org/variant/13-20763612-C-T), which is unusually high for a pathogenic variant. However, a ClinGen Hearing Loss expert panel has classified this variant as pathogenic for autosomal recessive nonsyndromic hearing loss, noting hearing loss is typically mild with incomplete and possibly age-related penetrance (https://www.ncbi.nlm.nih.gov/clinvar/variation/17023/; Shen et al. 2019. PubMed ID: 31160754). This variant is interpreted as pathogenic.

Pathogenic variants in *GJB2* have been primarily associated with autosomal recessive nonsyndromic deafness (OMIM #220290), including missense, nonsense, splicing, small insertions/deletions, and large deletions, all thought to represent loss of function alleles. Additionally, a small number of missense variants thought to represent gain of function alleles have been associated with autosomal dominant nonsyndromic deafness (OMIM #601544), as well as several autosomal dominant syndromic hearing loss disorders with mild to severe

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ectodermal defects variably involving the hair, skin and nails (Bart-Pumphrey syndrome (OMIM #149200), hystrix-like ichthyosis with deafness (OMIM #602540), keratitis-ichthyosis-deafness syndrome (OMIM #148210), palmoplantar keratoderma with deafness (OMIM #148350), and Vohwinkel syndrome (OMIM #124500).

This patient is apparently negative for clinically relevant copy number variants (CNVs). This patient is also negative for known microdeletion and microduplication syndromes, autosomal and sex chromosome aneuploidy, and triploidy.

These results should be interpreted in the context of clinical findings, family history and other laboratory data.

All genetic tests have limitations. Please see limitations and other information for this test on the following pages.

NOTES:

1) Based on these findings, the sequence variants considered in this patient are not eligible for free family follow up testing. Copy number variants are not eligible for free family follow up testing. For more information, please visit https://www.preventiongenetics.com/About/Resources/familytesting.

2) Reanalysis of this patient's sequence data at a later date (> 1 year later) may yield additional or new diagnoses (Wenger et al. 2016. PubMed ID: 27441994; Salfati et al. 2019. PubMed ID: 31847883).

3) Genetic counseling is recommended.

SUMMARY STATISTICS:

Pipeline	Version	Average NGS Coverage	Fraction Bases Covered with NGS
Infinity_Pipeline	4.0.0	172x	97.6%

Minimum NGS coverage is ≥20x for all coding exons and +/-10bp of flanking DNA.

Electronically signed on by:	Electronically signed and reported on by:

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SUPPLEMENTAL INFORMATION V.23.11

Background: PGxome is PreventionGenetics' whole exome sequencing (WES) assay. The PGxome assesses almost all genes from the human genome including coding regions and adjacent introns. This test is intended for health care providers who are looking for a genetic diagnosis. This is important as more than 50% of patients with genetic diseases are not given a specific diagnosis even after repeat clinical examinations and tests (Shashi et al. 2014). The standard clinical practice often involves examinations for specific phenotypes, imaging, biochemical testing for inborn errors of metabolism, genomic tests such as karyotyping or microarrays, and single gene or panel tests (Iglesias et al. 2014). However, patients remain without a genetic diagnosis, and patients and health care providers are caught in a long term diagnosis search, known as a diagnostic odyssey. This can lead to failures in identifying potential treatments, and unknown recurrence and prognosis risks (Yang et al. 2013).

Reported diagnostic rates from commercial and academic laboratories have found that WES assays have a ~20-40% positive diagnostic rate, with higher rates being reported from trio analysis (i.e. proband and parents) compared to singleton analysis (Atwal et al. 2014; Iglesias et al. 2014; Farwell et al. 2015). Notably, ~5-7% of individuals who have WES have had dual diagnoses (i.e. two non-overlapping clinical presentations) (Yang et al. 2014; Farwell et al. 2015; Posey et al. 2016). One study reported that 30% of genetics diagnoses have only been recently resolved due to new literature reports, highlighting the fast pace of gene-disease discovery and the need of genetic testing laboratories to be current of the medical literature (Yang et al. 2014). The use of a whole exome sequencing test may aid in altering clinical management, predict recurrence and prognosis risks, and reduce costs of additional testing, and may offer advantages over traditional molecular tests in certain patients (Valencia et al. 2015, Stark et al. 2017).

The PGxome is ideal for individuals with:

- Unresolved genetic testing such as normal karyotyping or microarray analysis, and negative single gene or gene panel sequencing results
- Disorders with significant genetic heterogeneity
- Global developmental delay/intellectual disability, with or without dysmorphic features
- Dysmorphic features, multiple congenital anomalies, or birth defects

Test Methods: For the PGxome we use Next Generation Sequencing (NGS) technologies to cover the coding regions of targeted genes plus ~10 bases of non-coding DNA flanking each exon. As required, genomic DNA is extracted from patient specimens. Patient DNA corresponding to these regions is captured using hybridization probes. Captured DNA is sequenced on the NovaSeq 6000 using 2x150 bp paired-end reads (Illumina, San Diego, CA, USA). The following quality control metrics are generally achieved: >97% of target bases are covered at >20x and mean coverage of target bases >100x. Data analysis and interpretation is performed by the internally developed Infinity pipeline. Variant calls are made by the GATK Haplotype caller and annotated using in house software and Jannovar. Common benign, likely benign, and low quality variants are filtered from analysis.

Copy number variants (CNVs) are also detected from NGS data. We utilize a CNV calling algorithm that compares mean read depth and distribution for each target in the test sample against multiple matched controls. Neighboring target read depth and distribution and zygosity of any variants within each target region are used to reinforce CNV calls.

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Reporting: Reports will consist of up to six different sections:

Primary Findings (related to the indication for testing)

- Variants in genes known to be associated with phenotype
- Variants in genes possibly associated with phenotype

Secondary Findings (if opted in on the requisition form)

- **Guideline Recommended Genes:** Recent recommendations are for labs performing WES or WGS to report pathogenic variants in selected medically actionable genes that cause (mostly) dominantly inherited disorders (v3.2; Miller et al. 2023. PubMed ID: 37347242).
- **Other Predispositions/Diagnoses:** This secondary finding option refers to a very broad range of disorders beyond the Recommended Genes above. Variants that are likely to result in a Mendelian (single gene) disorder will be reported (i.e., one variant in a dominant gene or X-linked gene or two variants in a recessive gene).
- **Carrier Status:** Variants in any gene that relate to an autosomal recessive or X-linked recessive (in females) disorder will be reported if this option is selected.
- **PG Discovery (Candidate Genes, Available for Trios Only):** WES provides the opportunity to identify rare variants in candidate genes for which there is limited available evidence. Relevant rare homozygous, hemizygous, compound heterozygous, and/or de novo variants are reported. These variants will not be confirmed by a second method (usually Sanger sequencing). Further research is required to understand if any human disease association exists. PreventionGenetics may reach out to request consent for submission of these variants to research programs and databases like GeneMatcher (https://genematcher.org/).

All differences from the reference sequences (sequence variants) are assigned to one of five interpretation categories (Pathogenic, Likely Pathogenic, Variant of Uncertain Significance, Likely Benign and Benign) per ACMG Guidelines (Richards et al. 2015). Pathogenic, Likely Pathogenic and Variants of Uncertain Significance will be reported in the first two sections. The remaining four sections are optional secondary findings categories that require an opt-in by the client in order for variants to be reported. For the third, fourth and fifth sections, only definitive pathogenic and likely pathogenic variants will be reported. For the last section, rare inherited or de novo variants of uncertain significance will be reported. Rare and undocumented synonymous variants are nearly always classified as likely benign if there is no indication that they alter protein sequence or disrupt splicing. Likely benign and benign variants are not included in the reports. However, a full list of all sequence variants will be provided to the ordering physician upon request.

Human Genome Variation Society (HGVS) recommendations are used to describe sequence variants (http://www.hgvs.org).

Limitations and Other Test Notes: Interpretation of the test results is limited by the information that is currently available. Better interpretation should be possible in the future as more data and knowledge about human genetics and this specific disorder are accumulated.

Sequencing: When sequencing does not reveal any heterozygous differences from the reference sequence, we cannot be certain that we were able to detect both patient alleles. Occasionally, a patient may carry an allele

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which does not capture or amplify, due for example to a large deletion or insertion. In these cases, the report may contain no information about the second allele.

For technical reasons, the PGxome test is not 100% sensitive. Some exons cannot be efficiently captured, and some genes cannot be accurately sequenced because of the presence of multiple copies in the genome. Therefore, a small fraction of sequence variants relevant to the patient's health will not be detected.

We sequence coding exons for most given transcripts, plus ~10 bp of flanking non-coding DNA for each exon. Unless specifically indicated, test reports contain no information about other portions of the gene, such as regulatory domains, deep intronic regions, uncharacterized alternative exons, chromosomal rearrangements, repeat expansions, and mitochondrial genome variants.

In most cases, we are unable to determine the phase of sequence variants. In particular, when we find two likely causative variants for recessive disorders, we cannot be certain that the variants are on different alleles.

The ability to detect low-level mosaicism of variants is limited.

Runs of mononucleotide repeats (eg (A)n or (T)n) with n >8 in the reference sequence are generally not analyzed because of strand slippage during amplification.

Unless otherwise indicated, DNA sequence data is obtained from a specific cell-type (usually leukocytes if taken from whole blood). Test reports contain no information about the DNA sequence in other cell-types.

We cannot be certain that the reference sequences are correct. Genome build hg19, GRCh37 (Feb2009) is used as our reference in nearly all cases.

Copy Number Variant Analysis: The PGxome test detects most deletions and duplications including intragenic CNVs and large cytogenetic events; however aberrations in a small percentage of regions may not be accurately detected due to sequence paralogy (e.g., pseudogenes, segmental duplications), sequence properties, deletion/duplication size (e.g., 1-3 exons vs. 4 or more exons), and inadequate coverage. In general, sensitivity for single, double, or triple exon CNVs is ~70% and for CNVs of four exon size or larger is >95%, but may vary from gene-to-gene based on exon size, depth of coverage, and characteristics of the region.

Balanced translocations or inversions are only rarely detected.

Certain types of sex chromosome aneuploidy may not be detected.

In nearly all cases, our ability to determine the exact copy number change within a targeted region is limited.

Our ability to detect CNVs due to somatic mosaicism is limited.

The sensitivity of this test is dependent on DNA quality.

General: We have confidence in our ability to track a specimen once it has been received by PreventionGenetics. However, we take no responsibility for any specimen labeling errors that occur before the specimen arrives at PreventionGenetics.

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A negative finding does not rule out a genetic diagnosis.

Genetic counseling to help to explain test results to the patients and to discuss reproductive options is recommended. Results of PGxome testing can be used for both diagnostic and scientific research purposes.

References: Please see the test description for this test on our website (www.preventiongenetics.com) for a full list of complete citations.

Regulatory Information: These results should be used in the context of available clinical findings, and should not be used as the sole basis for treatment. This test was developed and its performance characteristics determined by PreventionGenetics. US Food and Drug Administration (FDA) does not require this test to go through premarket FDA review. This laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA) as qualified to perform high complexity clinical laboratory testing.

If the results of this test will be utilized as part of a clinical study in the European Union, please note this test is intended for performance evaluation only. For all other testing within the European Union, this test is intended for research use only.

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